SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 2.1: Plant Nutrition — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Plant Nutrition
Driving Question	DRIVING QUESTION: Why does our sukuma wiki wilt in the midday sun even when the soil is wet — and what does a plant actually need to stay green, grow, and make its own food?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Wilting Mystery: Introducing the Anchoring Phenomenon**	Learners observed photographs and a short video clip of sukuma wiki plants wilting in a Nairobi shamba at midday, despite visibly moist soil. They also examined two potted plants side-by-side — one placed in full sun, one in shade — and recorded observable differences in leaf orientation, texture, and turgor. Learners noted their initial observations on sticky notes posted to the class Driving Question Board (DQB).	Learners established that wilting is not automatically caused by a lack of soil water, and that plants must be doing something active at midday that makes water availability matter even when the soil is wet. The class generated an initial list of 'what we think we know' and 'what we need to find out' about how plants get and use resources. Pedagogical note: Accept all student ideas at this stage without correction — the goal is to surface prior knowledge and misconceptions that subsequent lessons will address.	Learners cannot yet fully explain the phenomenon, but they articulated a productive initial question: if water is present in the soil, why is the plant losing it faster than it can absorb it, and what role does the sun play? This sets up the need to investigate leaf structure, photosynthesis, transpiration, and mineral nutrition across the unit. Teacher tip: Record every student-generated question on the DQB in the learner's own words — this creates ownership and gives you a formative baseline of class understanding.
Lesson 2 **Leaf Structure: The Architecture of a Photosynthetic Machine**	Learners prepared and examined wet-mount temporary slides of transverse sections of sukuma wiki (kale) leaves under a light microscope. They also studied pre-prepared slides and a detailed annotated diagram of a dicot leaf cross-section. Learners identified and sketched: the waxy cuticle, upper and lower epidermis, palisade mesophyll, spongy mesophyll, vascular bundles (xylem and phloem), guard cells, and stomata. They measured and compared the relative thickness of palisade	Each leaf layer is structurally adapted to a specific function: the transparent waxy cuticle and epidermis admit light while reducing water loss; the densely packed palisade mesophyll cells (rich in chloroplasts) are positioned to maximise light absorption for photosynthesis; the spongy mesophyll with its large air spaces facilitates gas exchange (CO_2 in, O_2 out); vascular bundles supply water and minerals via xylem and remove sugars via phloem; and guard cells control stomatal opening	Learners explained that the sukuma wiki leaf is not a passive surface — it is a highly organised structure that must simultaneously capture light, exchange gases, and manage water. The midday wilting phenomenon is now partially explained: when the sun is intense, the stomata may close to prevent water loss, but this also halts CO_2 entry and therefore photosynthesis. Learners updated the DQB with the new question: 'What exactly happens inside a chloroplast when light hits

	vs. spongy layers.	and closing, regulating both gas exchange and water loss. Pedagogical note: Explicitly connect stomata back to the wilting phenomenon — stomata must be open for CO_2 to enter (needed for photosynthesis), but an open stoma is also an exit for water vapour.	it?'
Lesson 3 **Chloroplasts and Pigments: Capturing the Sun's Energy**	Learners performed a paper chromatography investigation using crushed fresh sukuma wiki leaves and a suitable solvent (ethanol or petroleum ether). They separated leaf pigments on chromatography paper and measured Rf values for each pigment band. They identified at least four pigment bands: chlorophyll a (blue-green), chlorophyll b (yellow-green), carotenoids/xanthophylls (yellow/orange), and carotenes (orange). Learners also examined chloroplast structure from a detailed diagram and an electron micrograph, identifying the outer and inner membranes, stroma, grana (stacks of thylakoids), and thylakoid membranes.	Plants contain multiple photosynthetic pigments, each absorbing different wavelengths of visible light. Chlorophyll a is the primary pigment; chlorophyll b and carotenoids are accessory pigments that broaden the range of wavelengths absorbed and pass energy to chlorophyll a. This is why sukuma wiki leaves appear green — green wavelengths are reflected rather than absorbed. The chloroplast's internal architecture (grana with thylakoid membranes for light reactions; stroma for the Calvin cycle) is directly linked to the two stages of photosynthesis. Pedagogical note: Use the Rf calculation as a built-in maths integration opportunity. Have learners calculate Rf = distance moved by pigment ÷ distance moved by solvent front.	Learners explained that a plant needs multiple pigments to capture as much solar energy as possible across the visible spectrum — a single pigment would waste most of the available light. This connects to the driving question: the more light energy a plant captures efficiently, the more photosynthesis it can perform and the more it can keep cells turgid. Learners added 'What happens to the light energy once it is absorbed by chlorophyll?' to the DQB.
Lesson 4 **The Overall Equation of Photosynthesis — Evidence and Meaning**	Learners conducted the classic aquatic plant investigation using pondweed (or, in a Kenyan context, Elodea or locally sourced aquatic plants from a pond near school/Lake Victoria). Sprigs were placed in water under a bright lamp; learners counted oxygen bubbles produced per minute as a proxy for photosynthesis rate. They tested the effect of: (a) covering the beaker with black paper (dark), (b) normal room light, and (c) a lamp placed close to the beaker. Learners also reviewed a historical data card summarising Jan Ingenhousz's experiments and Van Helmont's willow tree experiment, contextualised with the work of modern plant biochemists.	Photosynthesis can be summarised by the overall equation: $6CO_2 + 6H_2O$ + light energy $\rightarrow C_6H_{12}O_6 + 6O_2$. Learners learned that: (1) light is essential — no bubbles were produced in darkness; (2) both CO_2 and water are raw materials; (3) glucose and oxygen are products; and (4) the process occurs in chloroplasts. The equation is a summary of many complex reactions, not a single step. Pedagogical note: Emphasise that the equation is a 'headline summary' — subsequent lessons (5 and 6) will unpack what actually happens inside the chloroplast in two distinct stages.	Learners explained that the sukuma wiki plant needs light, CO_2 (from the air via stomata), and water (from the soil via roots and xylem) to make its own glucose. This directly addresses the driving question: a plant needs all three inputs simultaneously. At midday, even if water and CO_2 are available, excessive heat may close stomata (limiting CO_2) or denature enzymes, reducing photosynthesis. Learners updated the DQB: 'How does the plant actually split water and fix CO_2 — what are the two stages?'
Lesson 5 **Light-Dependent Reactions: Where the Sun's Energy Enters the**	Learners watched an annotated animation (or teacher-drawn step-by-step diagram on the board) showing the light-dependent reactions on the thylakoid membrane. They	Learners traced the sequence of light-dependent reactions: light absorption by chlorophyll in PSII and PSI → excitation and transfer of electrons → photolysis of	Learners explained that the light-dependent reactions convert solar energy into chemical energy in the form of ATP and NADPH — the 'energy currency' the plant will spend in

Plant	traced the journey of: (1) a photon hitting Photosystem II → exciting electrons; (2) water molecules being split (photolysis) → releasing O_2, H^+, and electrons; (3) electrons passing along the electron transport chain → pumping H^+ to synthesise ATP (via ATP synthase); (4) electrons arriving at Photosystem I → re-energised by light → used to reduce $NADP^+$ to NADPH. Learners completed a sequencing card activity placing the steps in order.	water (source of O_2 released by the plant) → synthesis of ATP (via chemiosmosis across the thylakoid membrane) → reduction of $NADP^+$ to NADPH. The products ATP and NADPH are energy carriers that will power the Calvin cycle in the stroma. O_2 is a by-product released through stomata. Pedagogical note: Many learners incorrectly think the O_2 released by plants comes from CO_2 — explicitly address this misconception. The oxygen comes from the splitting of WATER, not CO_2. Isotope-labelling evidence (^{18}O) can be cited.	the next stage. This connects to the driving question: without adequate light (or with stomata closed, limiting water supply to mesophyll), the plant cannot generate sufficient ATP and NADPH, directly impairing its ability to grow and stay green. Learners added 'What does the plant do with ATP and NADPH?' to the DQB.
Lesson 6 **Light-Independent Reactions: How the Calvin Cycle Builds Glucose from CO_2**	Learners studied an annotated diagram of the Calvin cycle (stroma of chloroplast), tracing the three stages: (1) carbon fixation — CO_2 + RuBP (5C) → 2 × GP (3C), catalysed by RuBisCO; (2) reduction — GP is reduced using ATP and NADPH to form G3P (triose phosphate); (3) regeneration of RuBP — some G3P molecules are used to regenerate RuBP (using ATP), while others exit the cycle to form glucose, sucrose, starch, amino acids, and lipids. Learners used colour-coded molecular model kits (or drawn card cut-outs) to physically track carbon atoms through one turn of the cycle.	Learners learned that the light-independent reactions (Calvin cycle) occur in the stroma of the chloroplast, where CO_2 is fixed onto RuBP by RuBisCO to ultimately produce G3P (triose phosphate), which is used to synthesise glucose and other organic molecules. The cycle requires the ATP and NADPH produced in the light-dependent reactions — it is 'light-independent' because light is not used directly, but it cannot run without the products of the light reactions. For every 1 molecule of glucose produced, 6 turns of the cycle are needed, consuming 18 ATP and 12 NADPH. Pedagogical note: Stress that 'light-independent' does NOT mean 'happens in the dark' — a common misconception. Emphasise that RuBisCO is the most abundant enzyme on Earth and is essential to virtually all food chains, including the maize and beans Kenyan farmers grow.	Learners explained the full two-stage pathway of photosynthesis: light energy → ATP + NADPH (thylakoid membrane) → glucose (stroma). This completes the mechanistic answer to how sukuma wiki makes its own food. The driving question is now substantially answered for the photosynthesis component: the plant needs light, water, and CO_2 as inputs, and the two-stage process inside the chloroplast converts these into glucose that fuels growth, turgor, and all metabolic activity. Learners revised their DQB models.
Lesson 7 **Factors Affecting the Rate of Photosynthesis — Planning & Carrying Out an Investigation**	Learners planned and carried out a controlled investigation into one factor affecting photosynthesis rate (assigned variable: light intensity, CO_2 concentration, or temperature), using the pondweed/aquatic plant bubble-counting method or a data-logger with an O_2 sensor. For light intensity, a lamp was moved to measured distances from the plant; for CO_2, sodium hydrogen carbonate (bicarbonate) was added to the water in varying concentrations; for temperature, the water	The rate of photosynthesis increases as light intensity increases, but only up to a maximum — above that point the rate-limiting curve flattens, indicating that another factor (CO_2 concentration or temperature) has become limiting. Similarly, increasing CO_2 concentration increases the rate until light or temperature becomes limiting. Temperature affects enzyme activity (RuBisCO and others): rate increases with temperature up to an optimum (~25–35°C for most plants), then	Learners explained why the sukuma wiki farmer cannot simply give a plant 'more of everything' — each factor can independently limit the overall rate. The wilting phenomenon is now re-examined: at midday, high temperature may push leaf temperature beyond the enzyme optimum, closed stomata reduce CO_2, and water stress reduces turgor and the supply of H_2O for photolysis — multiple limiting factors act simultaneously. Learners updated the DQB with quantitative questions about optimal

	bath temperature was varied. Learners wrote a full prediction with scientific reasoning BEFORE collecting data, then recorded results in a table and plotted a graph. Emphasis was placed on NGSS Science and Engineering Practice 3 (Planning and Carrying Out Investigations) and Practice 4 (Analysing and Interpreting Data).	decreases sharply as enzymes denature. Learners learned to distinguish between independent variable, dependent variable, and controlled variables, and to identify sources of error. Pedagogical note: Connect the temperature findings directly to the Kenya context — Nairobi midday temperatures in February can exceed 30°C; Kericho tea-growing highlands rarely exceed 25°C. Ask learners why tea grows better in the highlands.	growing conditions.
Lesson 8 **Factors Affecting Photosynthesis Rate — Data Analysis & Explain**	Learners analysed secondary data sets: (1) published light-response curves for sukuma wiki/kale at two CO_2 concentrations; (2) temperature vs. photosynthesis rate graphs showing optimum, increase, and denaturation zones; and (3) a graph showing the effect of CO_2 concentration at two temperatures. Learners also reviewed a real dataset from a Kenyan agricultural research station (KALRO) on how shade netting affects yield in sukuma wiki production. They answered structured analysis questions requiring them to read values, calculate rates of change, identify limiting factors at specific points on the graph, and suggest practical interventions.	The rate of photosynthesis is controlled by limiting factors — light intensity, CO_2 concentration, and temperature — and the factor in shortest supply (or furthest from optimum) determines the overall rate (Blackman's Law of Limiting Factors). Learners learned to read and interpret multi-variable graphs, calculate percentage change, and make evidence-based predictions. They also learned that in commercial Kenyan horticulture (e.g., greenhouse sukuma wiki farming around Nairobi), farmers manipulate CO_2, temperature, and light to maximise yield. Pedagogical note: This lesson is the data-analysis companion to Lesson 7. Ensure all learners can correctly identify the limiting factor at any given point on a light-response curve before moving to Lesson 9.	Learners explained how all three limiting factors interact simultaneously in a real Kenyan shamba. The sukuma wiki wilting scenario is now fully explained from a photosynthesis-rate perspective: at midday, high irradiance could theoretically support high photosynthesis rates, but stomatal closure (triggered by water stress) reduces CO_2 supply and makes CO_2 the actual limiting factor — meaning the high light intensity is effectively 'wasted.' Learners can now advise a farmer: water in the early morning (before stomata close), consider drip irrigation, or use shade netting at peak hours.
Lesson 9 **Mineral Nutrition: What Does the Soil Give the Plant?**	Learners set up or observed results of a water-culture (hydroponics) investigation: maize or bean seedlings grown in (a) complete mineral solution, (b) solution lacking nitrogen, (c) solution lacking magnesium, and (d) distilled water only. They recorded plant height, leaf colour (green vs. yellow/chlorotic), and general vigour after two weeks. Learners also examined photographs of nutrient-deficiency symptoms in Kenyan crops (yellowing leaves in nitrogen-deficient maize on Rift Valley farms; purple tints in phosphorus-deficient beans; brown leaf margins in potassium-deficient sukuma wiki) and matched symptoms to deficient	Plants require macronutrients (N, P, K, Ca, Mg, S) and micronutrients (Fe, Zn, Mn, etc.) absorbed from soil water through root hair cells primarily by active transport (requiring ATP). Nitrogen is essential for amino acid, protein, and chlorophyll synthesis — deficiency causes yellowing (chlorosis) of older leaves. Magnesium is the central atom of chlorophyll — deficiency causes interveinal chlorosis. Phosphorus is needed for ATP, DNA, and cell membrane synthesis — deficiency causes stunted growth and purple coloration. Potassium regulates stomatal opening/closing (guard cell function) — deficiency causes leaf scorch. Learners connected mineral	Learners explained that the soil gives the plant far more than just water — it provides the mineral ions that are structural components of photosynthetic machinery (chlorophyll requires Mg; enzymes require N; energy transfer requires P). A sukuma wiki plant in K^+-deficient soil would have impaired stomatal regulation, making it wilt more easily even with adequate water. This extends the answer to the driving question: the plant needs the right minerals in addition to light, water, and CO_2 to stay green, grow, and photosynthesise efficiently.

	minerals. Soil pH test strips were used to test local soil samples and connect soil chemistry to nutrient availability.	nutrition directly to photosynthesis: without Mg, no chlorophyll; without N, no RuBisCO enzyme; without K, stomata cannot open properly. Pedagogical note: Emphasise the connection between K^+ deficiency and guard cell malfunction — this ties mineral nutrition directly back to the driving question's wilting phenomenon.	
Lesson 10 **Synthesis and Model Building: Answering the Driving Question**	Learners revisited the original anchoring phenomenon — sukuma wiki wilting at midday in wet soil — armed with the full unit's knowledge. They worked in groups to construct a comprehensive annotated diagram (concept model) showing: (1) water and mineral absorption by roots; (2) transport of water via xylem to leaves; (3) leaf structure (stomata, mesophyll, chloroplasts); (4) light-dependent reactions in thylakoid membrane (inputs: light, H_2O; outputs: ATP, NADPH, O_2); (5) Calvin cycle in stroma (inputs: CO_2, ATP, NADPH; output: glucose/G3P); (6) role of minerals (Mg in chlorophyll, N in enzymes, K in guard cells); and (7) how midday heat and high transpiration rate cause stomatal closure → CO_2 limitation → photosynthesis rate drops → glucose supply falls → turgor loss → wilting. Groups presented their models to the class for peer critique.	Learners synthesised all unit content into a unified mechanistic explanation: a plant stays green and grows by integrating water absorption, mineral nutrition, leaf architecture, pigment capture, light-dependent reactions, and the Calvin cycle — all of which must function simultaneously. Wilting in wet soil at midday is a systems-level response: water loss via transpiration exceeds uptake rate due to high evaporative demand, stomata close to conserve water, CO_2 entry is blocked, photosynthesis slows, sugar production drops, and cells lose turgor. This is not a single-cause failure — it is an emergent outcome of the interaction between multiple plant systems. Pedagogical note: Use the peer presentation of models as a summative formative assessment. Use a rubric assessing: accuracy of scientific content, correct use of terminology, quality of connections between concepts, and ability to answer peer questions. Celebrate the journey from 'I thought it needed more water' to a full mechanistic model.	Learners fully explained the driving question: sukuma wiki wilts at midday in wet soil because the rate of water loss (transpiration, driven by heat and vapour pressure gradient) exceeds the rate of water uptake, causing guard cells to close stomata. Closed stomata restrict CO_2 entry, making CO_2 a limiting factor for the Calvin cycle even when light intensity is high. Without sufficient Calvin cycle activity, glucose production falls, cells cannot maintain turgor, and the plant wilts — even with water in the soil. The solution for Kenyan small-scale farmers includes: morning watering, drip irrigation, shade netting during peak hours (10 am–2 pm), ensuring adequate K and Mg in soil, and choosing drought-tolerant sukuma wiki varieties. Learners posted their final models on the DQB alongside their Lesson 1 original questions, visually demonstrating their learning growth across the unit.